



(ii) Determine the linear attenuation coefficient μ of material Q.

$\mu = \dots\dots\dots \text{cm}^{-1}$ [3]

(iii) Use the information in (b)(i) to suggest why the X-rays emerging from the structure form an image that has poor contrast.

.....
.....
..... [1]

(c) Explain how X-rays are used in computed tomography (CT) scanning to produce a three-dimensional image of an internal structure.

.....
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.....
.....
..... [3]

[Total: 9]

